# Type MS · Turn a flowchart or pseudocode into Python

## Answer 1 of 8 · Question 3 part c · 10 marks

2023 JUN · Paper 02 · extract 1 of 6 · find part c in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 3(a) | A1<br>A2 | Award **one** mark for any of the following up to a maximum of **two** marks:<br>• Each column represents a variable/output (1)<br>• Rows shows how the values of variables change (as the code is executed) (1)<br>• Values are filled in manually/by hand (1) | Line numbers are for guidance – mark answer as a whole | **2** |

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|
| 3(b) | B1<br>B2<br>B3 | Award **one** mark for each correct cell:<br><br>**num1** \| **num2** \| **Output**<br>88 \| 18 \| State 2<br>17 \| 18 \| State 4<br>12 \| 19 \| State 1 | ignore quotes e.g. "State 2" | **3** |

## Type MS · Turn a flowchart or pseudocode into Python

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| Question | mp | Answer | Additional guidance | Mark |
|---|---|---|---|---|
| 3(c) | Award **one** mark for each of: | | C4 should exit loop<br><br>Allow C5 if a loop is used to keep asking for a non-negative exponent<br>C9 – ignore error message in C6; must have base, exponent, and answer in output for this mark | **10** |
| | C1 | At least one variable with a suitable variable name | | |
| | C2 | At least one variable initialised to an appropriate value | | |
| | C3 | While loop used to continue running code | | |
| | C4 | Relational test to identify terminating condition for loop | | |
| | C5 | Selection used to check for negative exponent | | |
| | C6 | Appropriate error message if exponent is negative | | |
| | C7 | Calculation of answer using any method | | |
| | C8 | A loop used to calculate answer as indicated in flowchart | | |
| | C9 | Prompts and output messages are fit for purpose | | |
| | C10 | Executing and producing the correct output for anticipated inputs | | |

| Suggested test data and output | | |
|---|---|---|
| **Base** | **Exponent** | **Output** |
| 0 | | ends program |
| 2 | 0 | 2 to power of 0 is 1 |
| 3 | 4 | 3 to power of 4 is 81 |

Type MS · Turn a flowchart or pseudocode into Python

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**C#**

```
// Initialise variables
int baseNum = 0;
int exponent = 0;
int answer = 1;
String outString = "";

// Display the first prompt and get the base number
Console.Write("Enter a base number (0 to exit)): ");
baseNum = Convert.ToInt32(Console.ReadLine());

// Check if user wants to exit
while (baseNum != 0.0)
{
    // Display a prompt and get the exponent number
    Console.Write("Enter an exponent number: ");
    exponent = Convert.ToInt32(Console.ReadLine());

    // Check if the exponent is negative
    if (exponent < 0)
    {
        Console.Write("Invalid exponent entered");
    }
    else
    {
        answer = 1;

        // Calculate the answer using exponentiation
        for (int i = 0; i < exponent; i++)
        {
            answer = answer * baseNum;
        }

        // Create a string for the output and show the user
        outString = baseNum.ToString() + " to the power of " +
                    exponent.ToString() + " is " +
                    answer.ToString();
        Console.Write(outString);
    }
    // Display the first prompt again and get the base number
    Console.Write("\nEnter a base number (0 to exit)): ");
    baseNum = Convert.ToInt32(Console.ReadLine());
}
```

# Type MS · Turn a flowchart or pseudocode into Python

## Answer 1 of 8 · Question 3 part c · 10 marks

2023 JUN · Paper 02 · extract 4 of 6 · find part c in this mark scheme extract

**Java**

```
// --------------------------------------------------
// Write your code below this line

// Initialise variables
int baseNum = 0;
int exponent = 0;
int answer = 1;
String outString = "";

Scanner myScanner = new Scanner (System.in);

// Display the first prompt and get the base number
System.out.print("Enter a base number (0 to exit)): ");
baseNum = Integer.parseInt (myScanner.nextLine());

// Check if user wants to exit
while (baseNum != 0.0)
{
    // Display a prompt and get the exponent number
    System.out.print("Enter an exponent number: ");
    exponent = Integer.parseInt (myScanner.nextLine());

    // Check if the exponent is negative
    if (exponent < 0)
    {
        System.out.println ("Invalid exponent entered");
    }
    else
    {
        answer = 1;

        // Calculate the answer using exponentiation
        for (int i=0; i<exponent; i++)
        {
            answer = answer * baseNum;
        }

        // Create a string for the output and show the user
        outString = Integer.toString(baseNum) + " to the power of " +
                    Integer.toString(exponent) + " is " +
                    Integer.toString(answer);
        System.out.println(outString);
    }

    // Display the first prompt again and get the base number
    System.out.print("Enter a base number (0 to exit)): ");
    baseNum = Integer.parseInt(myScanner.nextLine());
}
```

Type MS · Turn a flowchart or pseudocode into Python

Answer 1 of 8 · Question 3 part c · 10 marks

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**Python**

```
# Initialise variables
base = 0
exponent = 0
answer = 1

# Get the first base number
base = int (input ("Enter a base number (0 to exit): "))

# Check if the user wants to exit
while (base != 0):

    # Get the exponent number
    exponent = int (input ("Enter an exponent number: "))

    # Check if the exponent is negative
    if (exponent < 0):
        print ("Invalid exponent entered")
    else:
        # Reset the answer
        answer = 1

        # Calculate the answer using exponentiation
        for count in range (0, exponent):
            answer = answer * base

        # Create a string for the output sentence
        print (str (base) + " to the power of " +
               str (exponent) + " is " +
               str (answer))

    # Get another base number and lp
    base = int (input ("Enter a base number (0 to exit): "))
```

## Type MS · Turn a flowchart or pseudocode into Python

### Answer 1 of 8 · Question 3 part c · 10 marks

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| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|

## Type MS · Turn a flowchart or pseudocode into Python

### Answer 2 of 8 · Question 2 part a · 10 marks

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| | | | |
|---|---|---|---|
| 2(b)(i) | 24.0 (1) | Accept 24 | (1) |

## Type MS · Turn a flowchart or pseudocode into Python

### Answer 2 of 8 · Question 2 part a · 10 marks

2019 JUN · Paper 02 · extract 2 of 13 · find part a in this mark scheme extract

| | | | |
|---|---|---|---|
| 2(b)(ii) | One of:<br>By the rules of precedence multiplication is done before addition. (1)<br>Without the brackets the addition would be done to the total of the multiplication. (1)<br>By the rules of the BODMAS/BIDMAS additions happen after multiplication. (1)<br>newSpeed and intialSpeed have to be added together before multiplying (by time). (1) | | (1) |

## Type MS · Turn a flowchart or pseudocode into Python

### Answer 2 of 8 · Question 2 part a · 10 marks

2019 JUN · Paper 02 · extract 3 of 13 · find part a in this mark scheme extract

| Question | Answer | Additional Guidance | Mark |
|---|---|---|---|

## Type MS · Turn a flowchart or pseudocode into Python

### Answer 2 of 8 · Question 2 part a · 10 marks

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| | | | |
|---|---|---|---|
| 2(b)(iii) | Real (1)<br>Because real x integer = real (1) | Accept double / float<br>Accept 'because it could be a number with decimal places' | (2) |

## Type MS · Turn a flowchart or pseudocode into Python

### Answer 2 of 8 · Question 2 part a · 10 marks

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| | | | |
|---|---|---|---|
| 2(b)(iv) | 27.375 (1) | | (1) |

## Type MS · Turn a flowchart or pseudocode into Python

### Answer 2 of 8 · Question 2 part a · 10 marks

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| | | | |
|---|---|---|---|
| Q2(c)(i) | C#: 16 or 20 (1)<br>Java: 16 or 21 (1)<br>Python: 8 or 10 (1) | | (1) |

## Type MS · Turn a flowchart or pseudocode into Python

### Answer 2 of 8 · Question 2 part a · 10 marks

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| | | | |
|---|---|---|---|
| Q2(c)(ii) | C#: 29 or 31 (1)<br>Java: 24 or 26 (1)<br>Python: 14 or 15 (1) | | (1) |

## Type MS · Turn a flowchart or pseudocode into Python

### Answer 2 of 8 · Question 2 part a · 10 marks

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| | | | |
|---|---|---|---|
| Q2(c)(iii) | Any of:<br>1. Message (1)<br>2. startPos (Python) / pos (Java or C#) (1)<br>3. characters (1) | | (1) |

## Type MS · Turn a flowchart or pseudocode into Python

### Answer 2 of 8 · Question 2 part a · 10 marks

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| | | | |
|---|---|---|---|
| Q2(c)(iv) | newMessage (1) | Do not accept outMessage | (1) |

## Type MS · Turn a flowchart or pseudocode into Python

### Answer 2 of 8 · Question 2 part a · 10 marks

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| | | | |
|---|---|---|---|
| Q2(c)(v) | Response starts with 'c' (1)<br>Response has six letters in sequence from 'educational' (1) | Marks can be awarded from Q2(c)(vi) | (2) |

## Type MS · Turn a flowchart or pseudocode into Python

### Answer 2 of 8 · Question 2 part a · 10 marks

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| | | | |
|---|---|---|---|
| Q2(c)(vi) | 1. Returns substring / crops a string (1)<br>2. Located between the two input values / with number of characters identified in the last parameter (characters) and starting at the second parameter (startPos) (1) | Award two marks for:<br>It produces a substring of the message starting at position startPos with length characters<br>Award up to 2 marks for a correct interpretation of the code (e.g 'obtain a newMessage' – is worth 1)<br>Accept a correct generic explanation of what a subprogram does for 1 mark | (2) |

## Type MS · Turn a flowchart or pseudocode into Python

### Answer 2 of 8 · Question 2 part a · 10 marks

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| Question | Answer | Additional Guidance | Mark |
|---|---|---|---|

## Type MS · Turn a flowchart or pseudocode into Python

### Answer 2 of 8 · Question 2 part a · 10 marks

2019 JUN · Paper 02 · extract 13 of 13 · find part a in this mark scheme extract

| | | | |
|---|---|---|---|
| 2(a) | Award 1 mark for each of:<br>1. set counter to 1 and guess to 0 (1)<br>2. set answer to random value between 1 and 10 (1)<br>3. request input of and accept an integer value for guess (1)<br>4. create a while loop with a correct condition (1)<br>5. increment the counter (1)<br>6. use if else selection to determine and display appropriate output message for incorrect guesses (1)<br>7. request input of and accept another integer value for a guess inside the loop (1)<br>8. display correct guess and count of guesses (1)<br>9. compiling without syntax errors (1)<br>10. coding meets all requirements of question (1) | Candidates are required to open the file Q02a in the code editor.<br>Amended code should be saved as Q02aFINISHED.<br>Logic of algorithm must be followed as set out.<br>Alternatives must address each point.<br>Do not penalise candidates who attempt more than the stated requirements. | (10) |

| Question | Answer | Additional Guidance | Mark |
|---|---|---|---|

# Type MS · Turn a flowchart or pseudocode into Python

## Answer 3 of 8 · Question 2 part a · 10 marks

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| | | | | | |
|---|---|---|---|---|---|
| 2 (a) | A1 | 2.3.4 | Set variable base to 50 or heightChk to true | Logic of algorithm must be followed as set out.<br>Alternatives must address each point.<br>Do not penalise candidates who attempt more than the stated requirements.<br>Don't penalise spelling mistakes and alternative wording of the output. | (10) |
| | A2 | 2.2.2 | Create WHILE DO loop | | |
| | A3 | 2.4.1 | Request input of height | | |
| | A4 | 2.2.2 | if statement checks if height >= 1 or <= 100 | | |
| | A5 | 2.2.2 | if statement checks if height is >= 1 and <=100 | | |
| | A6 | 2.2.2 | heightCheck set to false if condition is met | | |
| | A7 | 2.5.1 | Set area using given formula (0.5 x base x height) | | |
| | A8 | 2.4.1 | Display one of base, height and area with appropriate label or displays two of base, height and area without labels | | |
| | A9 | 2.4.1 | Display all of base, height and area with appropriate labels | | |
| | A10 | 1.1.6 | Executing and producing correct output | | |
| Code example | | | | | |
| Python | | | | | |

```python
#   Q02aFINISHED

#   Initialise variables

base = 50

#   Print prompt and take input from user

    heightChk = TRUE
    while heightChk:
        height = int(input("Enter the height (between 1 and 100): "))
        if (height >= 1 and height <= 100):
            heightChk = FALSE

#   Calculate area and print out values

    area = 0.5 * base * height

    print ("Base length of triangle is : ", base)
    print ("Height of triangle is : ", height)
    print ("Area of triangle is : ", area)
```

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## Type MS · Turn a flowchart or pseudocode into Python

### Answer 3 of 8 · Question 2 part a · 10 marks

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| Question | mp | ref | Answer | Additional Guidance | Mark |
|---|---|---|---|---|---|

## Type MS · Turn a flowchart or pseudocode into Python

### Answer 3 of 8 · Question 2 part a · 10 marks

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| | | | | | |
|---|---|---|---|---|---|
| 2 (b) | B1 | 2.2.1 | Meaningful variable names used (1) | The code example shown is one way of responding to the task. Other methods should be credited accordingly. | (7) |
| | B2 | 2.4.1 | Meaningful prompts for input (1) | | |
| | B3 | 2.4.1 | Input of length and width as whole numbers (1) | | |
| | B4 | 2.5.1 | Perimeter calculated (= 2 x length + 2 x width) (1) | | |
| | B5 | 2.5.1 | Gap deducted from perimeter (1) | | |
| | B6 | 2.4.1 | Meaningful output (1) | | |
| | B7 | 2.1.1 | Code is fit for purpose. Must have mark points 1-6 and be easy to read (1) | | |

Code example

Python

```
1   #   Q02bFINISHED
2
3   #   Request input
4
5   length = int(input("Enter the length: "))
6   width = int(input("Enter the width: "))
7
8   #   Calculate number of panels needed
9
10  panels = 2 * length + 2 * width - 4
11
12  #   Print out number of panels needed
13
14  print ("Number of panels needed: " , panels)
```

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## Type MS · Turn a flowchart or pseudocode into Python

### Answer 3 of 8 · Question 2 part a · 10 marks

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| Question | mp | ref | Answer | Additional Guidance | Mark |
|---|---|---|---|---|---|

# Type MS · Turn a flowchart or pseudocode into Python

## Answer 3 of 8 · Question 2 part a · 10 marks

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For Q6, the first 11 marks are for coding that matches requirements of task. The remaining 9 marks should be allocated on a best fit.

| Question | mp | ref | Answer | Additional Guidance | Mark |
|---|---|---|---|---|---|
| 6 | A1 | 2.2.1 | Initialise variable for total sales by member of staff (1) | Sales for each member of staff<br><br>George Taylor 28466<br>Fehlix Chayne 35250<br>Sumren Bergen 30186<br>Samira Beckle 37430<br>Nellie Bogart 20448<br>Cheryl Grouth 34383<br>Danuta Graunt 26421<br>Jaiden Deckle 21507<br>Jimran Caliks 32415<br>Deynar Derran 28590<br>==========<br><br>Total staff sales: 295096<br><br>Highest sales by: Samira Beckle with 37430<br><br>Second highest sales by: Fehlix Chayne with 35250<br>>>> | (11) |
| | A2 | 2.5.1 | Add succeeding months' sales to total sales (1) | | |
| | A3 | 2.1.6 | Display total sales by member of staff (1) | | |
| | A4 | 2.2.2 | Loop repeats for each member of staff (1) | | |
| | A5 | 2.2.1 | Running team total sales initialised (1) | | |
| | A6 | 2.2.2 | Add succeeding individual staff sales to team total (1) | | |
| | A7 | 2.1.6 | Display final running total (1) | | |
| | A8 | 2.1.6 | Staff member with highest individual sales identified (1) | | |
| | A9 | 2.1.6 | Staff member with second highest individual sales identified (1) | | |
| | A10 | 2.1.6 | Information about staff with highest sales displayed (1) | | |
| | A11 | 2.1.6 | Information about staff with second highest sales displayed (1) | | |

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## Type MS · Turn a flowchart or pseudocode into Python

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| Band 1 (1-3 marks) | Band 2 (4-6 marks) | Band 3 (7-9 marks) | Mark |
|---|---|---|---|
| Little attempt to decompose the problem into component parts | Some attempt to decompose the problem into component parts | The problem has been decomposed into component parts | |
| Some parts of the logic are clear and appropriate to the problem | Most parts of the logic are clear and mostly appropriate to the problem | The logic is clear and appropriate to the problem | |
| Some appropriate use and manipulation of data types, variables, data structures and program constructs | The use and manipulation of data types, variables and data structures and program constructs is mostly appropriate | The use and manipulation of data types, variables and data structures and program constructs is appropriate | |
| Parts of the code are clear and readable | Code is mostly clear and readable | Code is clear and readable | |
| Finished program will not be flexible enough with other data sets or input | Finished program will function with some but not all other data sets or input | Finished program could be used with other data sets or input | |
| The program meets some of the given requirements | The program meets most of the given requirements | The program fully meets the given requirements | (9) |

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# Type MS · Turn a flowchart or pseudocode into Python

## Answer 3 of 8 · Question 2 part a · 10 marks

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Code example

Python

```
allStaffTotal = 0
highestSales = 0
secondSales = 0
high = 0
second = 0

print ("Sales for each member of staff","\n\n")
for staff in staffSales:
    staffTotal = 0
    sales = 3
    while (sales < 9):
        staffTotal = staffTotal + staff[sales]
        sales = sales + 1
    print (staff[1],staff[2],staffTotal)
    if (staffTotal > secondSales):
        if (staffTotal > highestSales):
            highestSales = staffTotal
            high = staff
        else:
            secondSales = staffTotal
            second = staff
    allStaffTotal = allStaffTotal + staffTotal
print ("==========\n\nTotal sales for all staff: ", allStaffTotal)

print ("\n\nHighest sales by: ",high[1],high[2], " with ", highestSales)
print ("\n\nSecond highest sales by: ",second[1],second[2], " with ", secondSales)
```

## Type MS · Turn a flowchart or pseudocode into Python

### Answer 4 of 8 · Question 2 part a · 11 marks

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<table>
<tr><th>Question</th><th>Mp</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>2 (c)</td><td colspan="2">Award 1 mark for each set of test data.</td><td></td><td></td></tr>
<tr><td></td><td>C1<br>C2<br>C3</td><td>
<table>
<tr><th></th><th>Test data</th><th>Expected results</th></tr>
<tr><td>booksSold<br>profit</td><td>Either of<br>• booksSold = 4<br>• profit = 4</td><td>Poor performances this week</td></tr>
<tr><td>booksSold</td><td>5</td><td rowspan="2">Sales and profit are good this week</td></tr>
<tr><td>profit</td><td>10</td></tr>
<tr><td>booksSold</td><td>21</td><td rowspan="2">Sales and profit are excellent this week</td></tr>
<tr><td>profit</td><td>20</td></tr>
</table>
</td><td></td><td>(3)</td></tr>
</table>

## Type MS · Turn a flowchart or pseudocode into Python

### Answer 4 of 8 · Question 2 part a · 11 marks

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| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|

# Type MS · Turn a flowchart or pseudocode into Python

## Answer 4 of 8 · Question 2 part a · 11 marks

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| Question | | Answer | Additional Guidance | Mark |
|---|---|---|---|---|
| 2 (a) | Award one mark for each of: | | Logic of algorithm must be followed as set out.<br><br>Alternatives must address each point.<br><br>Do not penalise candidates who attempt more than the stated requirements.<br><br>Do not penalise spelling mistakes in the assignment of username and/or password<br><br>Do not penalise spelling mistakes and alternative wording of the output. | (11) |
| | A1 | At least one variable with a suitable variable name | | |
| | A2 | username = bard423 | | |
| | A3 | password = nX2934? | | |
| | A4 | Loop used | | |
| | A5 | Username or password entered | | |
| | A6 | Username or password stored in variable(s) | | |
| | A7 | At least one suitable input message | | |
| | A8 | Checks username and password | | |
| | A9 | Appropriate error message(s) displays | | |
| | A10 | Welcome message displayed | | |
| | A11 | Executing and producing correct output (must have A4) | | |
| Code example | | | | |
| Python | | | | |

```
# Initialise variables
username = "bard423"
password = "nX2934?"
count = 0
inputUsername = ""
inputPassword = ""

# Print promts, take and check input from user
while inputUsername != username or inputPassword != password:
    if count > 0:
        print("There is a problem with the login details. Try again")
    count = count + 1
    inputUsername = input("Enter your username ")
    inputPassword = input("Enter your password ")
print("Welcome")
```

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### Answer 4 of 8 · Question 2 part a · 11 marks

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## Type MS · Turn a flowchart or pseudocode into Python

### Answer 4 of 8 · Question 2 part a · 11 marks

2021 JUN · Paper 02 · extract 5 of 6 · find part a in this mark scheme extract

| Question | Mp | Answer | | Additional Guidance | Mark |
|---|---|---|---|---|---|
| 2 (b) | Award 1 mark for each correct condition. | | | Accept alternatives e.g. Line 11 booksSold <=4 etc. | (4) |
| | | **Condition** | **Output message** | | |
| | B1 | Number of books sold is under 5 or profit made is under 5 | Poor performance this week | | |
| | B2 | Number of books sold is over 20; profit made is at least 20 | Sales and profit are excellent this week | | |
| | B3 | Number of books sold is at least 5; profit made is at least 10 | Sales and profit are good this week | | |
| | B4 | All other inputs | Alert manager | | |
| Code example | | | | | |
| Python | | | | | |

```
11 if booksSold < 5 or profit < 5:
12     print("Poor performance this week")
13 elif booksSold > 20 and profit >= 20:
14     print("Sales and profit are excellent this week")
15 elif booksSold >=5 and profit >= 10:
16     print("Sales and profit are good this week")
17 else:
18     print("Alert manager")
```

## Type MS · Turn a flowchart or pseudocode into Python

### Answer 4 of 8 · Question 2 part a · 11 marks

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## Type MS · Turn a flowchart or pseudocode into Python

### Answer 5 of 8 · Question 2 part a · 10 marks

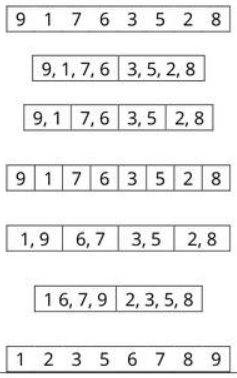
2021 NOV · Paper 02 · extract 1 of 18 · find part a in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 2 (b) | B1<br>B2<br>B3<br>B4 | **Award up to 4 marks for a linked explanation such as:**<br><br>• The list is not sorted (1) Juan should come after Elija if the list was sorted / Elija is compared with Juan (1), so the bottom half of the list would be discarded (1) after the first pass through the loop (1)<br>• Binary search works on a sorted list (1) because it uses divide and conquer (1) half the list discarded each pass through (1) based on comparison of search item with middle item (1) | | (4) |
| **Question** | **mp** | **Answer** | **Additional Guidance** | **Mark** |

## Type MS · Turn a flowchart or pseudocode into Python

### Answer 5 of 8 · Question 2 part a · 10 marks

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| | | | | |
|---|---|---|---|---|
| 2 (c) | C1<br>C2<br>C3<br>C4<br>C5 | **Award 1 mark for each of:**<br>Split into two sets of four (1)<br>Split each set into two pairs (1)<br>Split each pair into single elements (1)<br>Merge elements into sorted pairs (1)<br>Merge pairs into sorted sets (1)<br>  | Allow step at C3 to be implied | (5) |

## Type MS · Turn a flowchart or pseudocode into Python

### Answer 5 of 8 · Question 2 part a · 10 marks

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| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|

## Type MS · Turn a flowchart or pseudocode into Python

### Answer 5 of 8 · Question 2 part a · 10 marks

2021 NOV · Paper 02 · extract 4 of 18 · find part a in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 2 (a) | **Award one mark for each of:** | | Logic of algorithm must be followed as set out.<br>Alternatives must address each point.<br>Do not penalise candidates who attempt more than the stated requirements.<br>Do not penalise spelling mistakes in the input message. | |
| | A1 | Input as an integer (1) | | |
| | A2 | Loop with correct conditions<br>start <= end and not found (1) | | |
| | A3 | Use of correct symbol for integer division (in conversion of DIV) or type coercion (1) | | |
| | A4 | Check to see if number in middle matches item (1) | | |
| | A5 | Set found to true (1) | | |
| | A6 | Check to see if item is less than number in middle (1) | | |
| | A7 | Set end to middle – 1 (1) | | |
| | A8 | else – set start to middle + 1 (1) | | |
| | A9 | count incremented by 1 (1) | | |
| | A10 | Program execution is fully correct (1) | | (10) |
| Code examples | | | | |

## Type MS · Turn a flowchart or pseudocode into Python

### Answer 5 of 8 · Question 2 part a · 10 marks

2021 NOV · Paper 02 · extract 5 of 18 · find part a in this mark scheme extract

C#

```
Console.WriteLine("What is the number to find? ");
int item = Convert.ToInt32(Console.ReadLine());

while (start <= end && !found)
{
    middle = (start + end) / 2;
    if (numberList[middle] == item)
    {
        found = true;
    }
    else
    {
        if (item < numberList[middle])
        {
            end = middle - 1;
        }
        else
        {
            start = middle + 1;
        }
    }
    count++;
}
```

## Type MS · Turn a flowchart or pseudocode into Python

### Answer 5 of 8 · Question 2 part a · 10 marks

2021 NOV · Paper 02 · extract 6 of 18 · find part a in this mark scheme extract

Java

```
Scanner input = new Scanner(System.in);
System.out.print("What is the number to find? ");
int item = Integer.parseInt(input.nextLine());

while (start <= end && !found)
{
    middle = (start + end) / 2;
    if (numberList[middle] == item)
    {
        found = true;
    }
    else
    {
      if (item < numberList[middle])
      {
          end = middle - 1;
      }
      else
      {
          start = middle + 1;
      }
    }
    count ++;
}
```

Python

```
item = int(input("What is the number to find? "))
while start <= end and not found:
    middle = (start + end) // 2
    if numberList[middle] == item:
        found = True
    else:
        if item < numberList[middle]:
            end = middle - 1
        else:
            start = middle + 1
    count = count + 1
```

## Type MS · Turn a flowchart or pseudocode into Python

### Answer 5 of 8 · Question 2 part a · 10 marks

2021 NOV · Paper 02 · extract 7 of 18 · find part a in this mark scheme extract

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|

## Type MS · Turn a flowchart or pseudocode into Python

### Answer 5 of 8 · Question 2 part a · 10 marks

2021 NOV · Paper 02 · extract 8 of 18 · find part a in this mark scheme extract

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|
| 4 (a) | **Award one mark for each of:** | | | |
| | | **Validation** | | |
| | A1 | Suitable input prompt and storing it in binaryPattern (1) | | |
| | A2 | Length of binary pattern validated as 8 (1) | | |
| | A3 | Loop used to check the length of binary pattern (1) | | |
| | | **Binary to denary** | | |
| | A4 | Loop through each digit in the binary pattern (1) | | |
| | A5 | Correct conversion of at least one binary digit to relevant placeholder (1) | | |
| | A6 | Attempt at running total using denaryNumber (1) | | |
| | A7 | Correct running total using denaryNumber (1) | | |
| | A8 | Print statement includes the binary pattern and the denary number (1) | | |
| | A9 | Print statement outside of conversion loop (1) | | |
| | A10 | Comment explains how conversion works (1) | | |
| | A11 | Program executes correctly for any binary pattern that is exactly 8 characters long only (1) | | (11) |
| Code examples | | | | |

## Type MS · Turn a flowchart or pseudocode into Python

### Answer 5 of 8 · Question 2 part a · 10 marks

2021 NOV · Paper 02 · extract 9 of 18 · find part a in this mark scheme extract

C#

```
while (binaryPattern.Length != 8)
{
    Console.WriteLine("Enter an 8 digit binary number to convert to denary ");
    binaryPattern = Console.ReadLine();
}

foreach (char digit in binaryPattern)
{
    if (digit == '1')
    {
        denaryNumber += denaryPlaceholders[count] * 1;
    }
    count++;
}
Console.WriteLine(binaryPattern + " converted to denary is " + denaryNumber);
Console.ReadLine();
```

Java

```
while (binaryPattern.length() != 8)
{
    Scanner input = new Scanner(System.in);
    System.out.print("Enter an 8 digit binary number to convert to denary ");
    binaryPattern = input.next();
}

for (char digit : binaryPattern.toCharArray())
{
    if (digit == '1')
    {
       denaryNumber += denaryPlaceholders[count] * 1;
    }
    count ++;
}
System.out.print(binaryPattern + " converted to denary is " + denaryNumber);
```

## Type MS · Turn a flowchart or pseudocode into Python

### Answer 5 of 8 · Question 2 part a · 10 marks

2021 NOV · Paper 02 · extract 10 of 18 · find part a in this mark scheme extract

Python

```
while len(binaryPattern) != 8:
    binaryPattern = input("Enter an 8 digit binary number to convert to denary: ")

for digit in binaryPattern:
    if digit == "1":
        denaryNumber = denaryNumber + denaryPlaceholders[count] * 1
    count = count + 1

print(binaryPattern,"converted to denary is", denaryNumber)
```

# Type MS · Turn a flowchart or pseudocode into Python

## Answer 5 of 8 · Question 2 part a · 10 marks

2021 NOV · Paper 02 · extract 11 of 18 · find part a in this mark scheme extract

For Q5, the first 11 marks are for coding that matches requirements of task. The remaining 9 marks should be allocated on a best fit

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|
| 5 | A1 | At least three variables initialised appropriately | | (1) |
| | A2 | Word input | | (1) |
| | A3 | Program repeats until 1 is input | | (1) |
| | A4 | Loop through each word in the array | | (1) |
| | | Words that begin with the same letter | | |
| | A5 | Check to see if the first letter of the input word matches the first letter of at least one word in the array | | (1) |
| | A6 | The number of words that begin with the same letter displayed | | (1) |
| | | Words that contain the input word | | |
| | A7 | Check to see if the word contains the input word | | (1) |
| | A8 | Number of letters in the word calculated | | (1) |
| | A9 | At least one word from the array that contains the input word displayed | | (1) |
| | A10 | The number of words that contain the input word displayed | | (1) |
| | A11 | Identification of the longest word or the shortest word | | (1) |

| Band 1 (1-3 marks) | Band 2 (4-6 marks) | Band 3 (7-9 marks) | Mark |
|---|---|---|---|
| Little attempt to decompose into component parts | Some attempt to decompose into component parts | The problem has been decomposed into component parts | |
| Some parts of the logic are clear and appropriate to the problem | Most parts of the logic are clear and mostly appropriate to the problem | The logic is clear and appropriate to the problem | |
| Some appropriate use and manipulation of data types, variables, data structures and program constructs | The use and manipulation of data types, variables and data structures and program constructs is mostly appropriate | The use and manipulation of data types, variables and data structures and program constructs is appropriate | |
| Parts of the code are clear and readable | Code is mostly clear and readable | Code is clear and readable | |
| Finished program will not be flexible enough with other data sets or input | Finished program will function with some but not all other data sets or input | Finished program could be used with other data sets or input | |
| The program meets some of the given requirements | The program meets most of the given requirements | The program fully meets the given requirements | (9) |

# Type MS · Turn a flowchart or pseudocode into Python

## Answer 5 of 8 · Question 2 part a · 10 marks

2021 NOV · Paper 02 · extract 12 of 18 · find part a in this mark scheme extract

Code examples

C#

```
// Add your code here
string inputWord = "";

while (inputWord != "1")
{
    int shortest = 5000;
    int longest = 0;
    String shortestWord = "";
    String longestWord = "";

    Console.WriteLine("Enter a word or 1 to exit ");
    inputWord = Console.ReadLine();

    if (inputWord != "1")
    {
        int count = 0;

        foreach (String word in wordArray)
        {
            if(word[0] == inputWord[0])
            {
                Console.WriteLine(word);
                count ++;
            }
        } // End of loop

        Console.WriteLine(count + " word(s) beginning with "+ inputWord[0]);
        Console.WriteLine("----------------------------------------------------------------------");

        count = 0;

        foreach (String word in wordArray)
        {
            if(word.Contains(inputWord))
            {
                count ++;
                Console.WriteLine(word);

                int length = word.Length;

                if (longest < length)
```

# Type MS · Turn a flowchart or pseudocode into Python

## Answer 5 of 8 · Question 2 part a · 10 marks

2021 NOV · Paper 02 · extract 13 of 18 · find part a in this mark scheme extract



```
                    {
                        longest = length;
                        longestWord = word;
                    }

                    if (shortest > length)
                    {
                        shortest = length;
                        shortestWord = word;
                    }
                }// end of checking word
            } // End of for loop

            // Print the number of words, the number of characters in the longest and
            // shortest word, the longest and the shortest word

            if (count > 0)
            {
                Console.WriteLine(count + " word(s) with "+ inputWord + " in them");
                Console.WriteLine("The longest word has " + longest + " characters");
                Console.WriteLine("The shortest word has " + shortest + " characters");
                Console.WriteLine("The longest word is " + longestWord + " characters");
                Console.WriteLine("The shortest word is " + shortestWord + " characters");
                Console.WriteLine("--------------------------------------------------------------------------------");
            }
            else
            {
                Console.WriteLine("There were 0 words that had all the letters from " + inputWord + " in them");
                Console.WriteLine("--------------------------------------------------------------------------------");
            } // End of printing words
        } // End of check
    } // End of while loop
```

# Type MS · Turn a flowchart or pseudocode into Python

## Answer 5 of 8 · Question 2 part a · 10 marks

2021 NOV · Paper 02 · extract 14 of 18 · find part a in this mark scheme extract

Java

```java
// Add your code here
Scanner input = new Scanner(System.in);
String inputWord = "";

// While there is a word run the program
while (!inputWord.equals("1")) {

    int shortest = 5000;
    int longest = 0;
    String shortestWord = "";
    String longestWord = "";

    System.out.println("Enter a word or 1 to exit ");
    System.out.println("--------------------------------------------------------------------------------");
    inputWord = input.nextLine();
    if (!inputWord.equals("1")){
        // Find words that begin with the same letter as the input word
        int count = 0;

        // Get each word in the array
        for (String word: wordArray){

            // If the first character of the word is the same as the first character of
            // the input word then print the word
            if (word.charAt(0) == inputWord.charAt(0)){
                System.out.println(word);
                count ++;
            }
        } // End of for loop

        // Print the total number of words that begin with the same letter
        System.out.println(count + " word(s) beginning with "+ inputWord.charAt(0));
        System.out.println("--------------------------------------------------------------------------------");

        // Find words that contain the input word
        count = 0;

        // Get each word in the array
        for (String word: wordArray){
            // Check to see if the word contains the inputWord
            if (word.contains(inputWord)){
                count ++;
                System.out.println(word);

                // Set the length of the word
                int length = word.length();
```

# Type MS · Turn a flowchart or pseudocode into Python

## Answer 5 of 8 · Question 2 part a · 10 marks

2021 NOV · Paper 02 · extract 15 of 18 · find part a in this mark scheme extract



```
                    // If the length is longer than the current longest replace it and the longestWord
                    if (longest < length) {
                        longest = length;
                        longestWord = word;
                    }

                    // If the length is shorter than the shortest replace it and the shortestWord
                    if (shortest > length){
                        shortest = length;
                        shortestWord = word;
                    }
                } // End of checking word
            } // End of for loop

            // Print the number of words, the number of characters in the longest and
            // shortest word, the longest and the shortest word
            if( count > 0){
                System.out.println(count + " word(s) with "+ inputWord + " in them");
                System.out.println("The longest word has " + longest + " characters");
                System.out.println("The shortest word has " + shortest + " characters");
                System.out.println("The longest word is " + longestWord + " characters");
                System.out.println("The shortest word is " + shortestWord + " characters");
                System.out.println("--------------------------------------------------------------------------------");
            } else {
                System.out.println("There were 0 words that had all the letters from " + inputWord + " in them");
                System.out.println("--------------------------------------------------------------------------------");
            } // End of printing words
        }
    } // End of while loop
```

# Type MS · Turn a flowchart or pseudocode into Python

## Answer 5 of 8 · Question 2 part a · 10 marks

2021 NOV · Paper 02 · extract 16 of 18 · find part a in this mark scheme extract

Python

```
# While there is a word run the program
while inputWord !="1":

    shortest = 5000
    longest = 0
    shortestWord = ""
    longestWord = ""

    inputWord = input("Enter a word or 1 to exit: ")
    print("------------------------------------------------------------------------")

    if inputWord !="1":
        # Find words that begin with the same letter as the input word
        count = 0
        # Get each word in the array
        for word in wordArray:
            # If the first character of the word is the same as the first character of
            # the input word then print the word
            if word[0] == inputWord[0]:
                print(word)
                count = count + 1

        # Print the total number of words that begin with the same letter
        print("\n",count,"word(s) beginning with",inputWord[0])
        print("------------------------------------------------------------------------")

        # Find words that contain the input word
        count = 0
        # Get each word in the array
        for word in wordArray:
            # Check to see if the word contains the inputWord
            if inputWord in word:
                count = count + 1
                print(word)
                # Set the length of the word
                length = len(word)

                # If the length is longer than the current longest replace it and the longestWord
                if longest < length:
                    longest = length
                    longestWord = word

                # If the length is shorter than the shortest replace it and the shortestWord
                if shortest > length:
                    shortest = length
                    shortestWord = word
```

# Type MS · Turn a flowchart or pseudocode into Python

## Answer 5 of 8 · Question 2 part a · 10 marks

2021 NOV · Paper 02 · extract 17 of 18 · find part a in this mark scheme extract

```
59
60          # Print the number of words, the number of characters in the longest and
61          # shortest word, the longest and the shortest word
62          if count > 0:
63              print("\n",count,"word(s) with",inputWord,"in them")
64              print("The longest word has",longest,"letters")
65              print("The shortest word has",shortest,"letters")
66              print("The longest word is",longestWord)
67              print("The shortest word is",shortestWord)
68              print("--------------------------------------------------------------------------")
69          else:
70              print("There were 0 words with all the letters from",inputWord,"in them")
71          # ------------------------------------------------------------------------------------
72
73  # End the program when there are no more words
74  print("End of program")
```

## Type MS · Turn a flowchart or pseudocode into Python

### Answer 5 of 8 · Question 2 part a · 10 marks

2021 NOV · Paper 02 · extract 18 of 18 · find part a in this mark scheme extract

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## Type MS · Turn a flowchart or pseudocode into Python

### Answer 6 of 8 · Question 2 part b · 11 marks

2022 JUN · Paper 02 · extract 1 of 7 · find part b in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 2 (a)(i) | A1 | 3/8/20 | | (1) |

## Type MS · Turn a flowchart or pseudocode into Python

### Answer 6 of 8 · Question 2 part b · 11 marks

2022 JUN · Paper 02 · extract 2 of 7 · find part b in this mark scheme extract



## Type MS · Turn a flowchart or pseudocode into Python

### Answer 6 of 8 · Question 2 part b · 11 marks

2022 JUN · Paper 02 · extract 3 of 7 · find part b in this mark scheme extract



## Type MS · Turn a flowchart or pseudocode into Python

### Answer 6 of 8 · Question 2 part b · 11 marks

2022 JUN · Paper 02 · extract 4 of 7 · find part b in this mark scheme extract



## Type MS · Turn a flowchart or pseudocode into Python

### Answer 6 of 8 · Question 2 part b · 11 marks

2022 JUN · Paper 02 · extract 5 of 7 · find part b in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 2 (a)(v) | A5 | pNumber | | (1) |

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|

# Type MS · Turn a flowchart or pseudocode into Python

## Answer 6 of 8 · Question 2 part b · 11 marks

2022 JUN · Paper 02 · extract 6 of 7 · find part b in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 2 (b) | **Award 1 mark for each of:** | | Logic of algorithm must be followed as set out.<br>Alternatives must address each point.<br>Do not penalise candidates who attempt more than the stated requirements.<br>Do not penalise spelling mistakes in the input message. | (11) |
| | **Evidence should be found in the Function** | | | |
| | B1 | Function interface is correct (1) | | |
| | B2 | Condition correct (for the number 1) (1) | | |
| | B3 | Loop initialised correctly (from 2 to the input number) (1) | | |
| | B4 | If condition correct (modulus) (1) | | |
| | B5 | Check returned (1) | | |
| | **Evidence should be found in the Main program** | | | |
| | B6 | Input of number stored as an integer variable (1) | | |
| | B7 | Input includes a space/colon / new line before the input value | | |
| | B8 | Function called correctly (1) | | |
| | B9 | Display includes number and message for either a prime number **OR** not (1) | | |
| | **Overall** | | | |
| | B10 | Compiling without syntax errors (1) | | |
| | B11 | Executing and producing the correct output (1) | | |
| Code examples | | | | |
| Python | | | | |

```python
# Write the function here
def checkPrime(pNumber):
    if pNumber == 1:
        check = False
    else:
        check = True
        for count in range(2,pNumber):
            if pNumber % count == 0 :
                check = False
    return check

# Write the main program here
number = int(input("Enter a number: "))
result = checkPrime(number)
if result == True:
    print(number,"is a prime number")
else:
    print(number,"is not a prime number")
```

## Type MS · Turn a flowchart or pseudocode into Python

### Answer 6 of 8 · Question 2 part b · 11 marks

2022 JUN · Paper 02 · extract 7 of 7 · find part b in this mark scheme extract

| Question | mp | | Additional Guidance | Mark |
|---|---|---|---|---|

## Type MS · Turn a flowchart or pseudocode into Python

### Answer 7 of 8 · Question 3 part c · 5 marks

2025 JUN · Paper 02 · extract 1 of 5 · find part c in this mark scheme extract

| Question | Mp | Answer | Additional Guidance | Marks |
|---|---|---|---|---|
| 3(a)(i) | A1<br>A2 | Award **one** mark for each:<br>• Pass 1 correct (1)<br>• Pass 2 correct (1)<br>9 11 12 19 14 18 15 17<br>9 11 12 14 18 15 17 19 (1)<br>9 11 12 14 15 17 18 19 (1)<br><br>Must show ONLY the 19 moving in first pass and ONLY the 18 moving in second pass | Ignore lines with individual swaps and only award correct completed passes<br>9 11 12 19 14 18 15 17<br>9 11 12 **14** **19** 18 15 17<br>9 11 12 14 **18** **19** 15 17<br>9 11 12 14 18 **15** **19** 17<br>9 11 12 14 18 15 **17** **19** (1)<br>9 11 12 14 **15** **18** 17 19<br>9 11 12 14 15 **17** **18** 19 (1)<br><br>Accept **accurate** right-to-left bubble sort as shown below<br>9 11 12 19 14 18 15 17<br>9 11 12 14 19 15 18 17 (1)<br>9 11 12 14 15 19 17 18<br>9 11 12 14 15 17 19 18<br>9 11 12 14 18 15 17 19 (1)<br><br>Complete sort cannot be the 1st /only line in response – 0 marks as this will be taken as both 18 and 19 moving in first pass | **2** |

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# Type MS · Turn a flowchart or pseudocode into Python

## Answer 7 of 8 · Question 3 part c · 5 marks

2025 JUN · Paper 02 · extract 2 of 5 · find part c in this mark scheme extract

Paper 2 – Application of Computational Thinking

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|
| 3(a)(ii) | A3 | Award **one** mark for:<br>• 4 | If right-to-left pass completed in Q3ai, can only accept 2 swaps | **1** |

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|
| 3(a)(iii) | A4 | Award **one** mark for:<br>• 3 | Do not accept 2 – 3rd pass required for **algorithm** to finish<br>If right-to-left pass completed in Q3ai, can only accept 5 passes | **1** |

| Question | mp | Answer | Additional guidance | Mark |
|---|---|---|---|---|
| 3(b) | B1 | Award **one** mark for:<br>• Binary search | Accept any other divide and conquer search algorithm (google if necessary)<br>Accept 'binary' on its own as search in question | **2** |
| | B2 | Award **one** mark for any of the following:<br>• Divide and conquer / list halves each time(1)<br>• Fewer searches needed (1) | Do not accept points about being 'already sorted' as this is in question | |

10 of 23

# Type MS · Turn a flowchart or pseudocode into Python

## Answer 7 of 8 · Question 3 part c · 5 marks

2025 JUN · Paper 02 · extract 3 of 5 · find part c in this mark scheme extract

Paper 2 – Application of Computational Thinking

| Question | mp | Answer | Additional guidance | Mark |
|---|---|---|---|---|
| 3(c) | | Award **one** mark for each of: | | |
| | C1 | (while) Loop ends **only** when 0 entered (1) | Any method<br>e.g. `while age != 0:` | |
| | C2 | Avoiding divide by zero error (1) | By coding 2nd selection box from flowchart – no output required<br>Must come after some attempt to process age or count – award even if inside loop.<br>ignore 'other' divide by zero errors introduced by candidate<br>Must follow logic of flowchart e.g.:<br>`if count == 0: (or count != 0)`<br>…<br>`else`<br>… | |
| | C3 | Average value rounded to whole number (1) | Must come after some attempt to process age or count – award even if inside loop.<br>Must target calculated average (even if formula not correct)<br>Any method to get int<br>Accept casting to int *(round not in spec, but string manipulation is)*<br>Accept integer division | **5** |

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# Type MS · Turn a flowchart or pseudocode into Python

## Answer 7 of 8 · Question 3 part c · 5 marks

2025 JUN · Paper 02 · extract 4 of 5 · find part c in this mark scheme extract



| | | | | |
|---|---|---|---|---|
| | C4 | Correct value calculated for average and count (1) | Allow even if 0 used to initialise count in B1 (this must be only error in calculation) – set count to -1 to test | |
| | C5 | Count and average included in a syntactically correct final print statement (1) | Award even if never runs<br>Must include both variables (may also include string) | |

Correct

```
# Q03c

# program variables
age = 1
count = -1
total = 0

# ------------------------------------------------
# Write your code below this line

while age != 0:

    age = int(input("Please enter age in years: "))

    total = total + age
    count = count + 1

if count == 0:
    print("No ages input")
else:
    average = round(total / count)
    print("You entered", count, "ages; the average age is", average)
```

Typical

```
# Q03c

# program variables
age = 0
count = 0
total = 0

# ------------------------------------------------
# Write your code below this line
age = int(input("Please enter age in years: "))
while age != 0:

    total = total + age
    count = count + 1

    age = int(input("Please enter age in years: "))

if count == 0:
    print("No ages input")
else:
    average = int(total / count)
    print("You entered", count, "ages; the average age is", average)
```

## Type MS · Turn a flowchart or pseudocode into Python

### Answer 7 of 8 · Question 3 part c · 5 marks

2025 JUN · Paper 02 · extract 5 of 5 · find part c in this mark scheme extract

Paper 2 – Application of Computational Thinking

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|

# Type MS · Turn a flowchart or pseudocode into Python

## Answer 8 of 8 · Question 4 part c · 8 marks

2025 JUN · Paper 02 · extract 1 of 5 · find part c in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 4(a) | A1 | Breaking problem into small parts | | 1 |

| Question | mp | Answer | Additional guidance | Mark |
|---|---|---|---|---|
| 4(b) | | Award **two** marks for a linked explanation such as: | | |
| | B1<br>B2 | • Programmer can ignore unnecessary detail (1) because it doesn't matter how library generates the number (1)<br>• Eli can focus on just his own code (1) because library hides the code that generates the random numbers (1)<br>• It doesn't matter how the subprogram generates the random number (1) because Eli only needs the number itself (1) | First mark for meaning of abstraction (focus on necessary parts / hiding unnecessary detail)<br>Second mark for linking use of **random** library<br>Do **not** accept answers relating to modelling real-world scenarios as this is in the question | 2 |

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# Type MS · Turn a flowchart or pseudocode into Python

## Answer 8 of 8 · Question 4 part c · 8 marks

2025 JUN · Paper 02 · extract 2 of 5 · find part c in this mark scheme extract

Paper 2 – Application of Computational Thinking

| Question | mp | Answer | Additional guidance | Mark |
|---|---|---|---|---|
| 4(c) | | Award **one** mark for each of: | | |
| | C1 | Booster variable initialised to 1 (1) | booster = 1 | |
| | C2 | At least one given constant used correctly (1) | | |
| | C3 | Loop used to make three throws (1) | Either type | |
| | C4 | Use of random.randint( ) to assign roll a value between 1 and 6 (1) | roll = random.randint(1,SIDES)<br>roll = random.randint(1,6)<br>Any accurate method<br>do **not** accept:<br>random.randint(6) random.randint(SIDES) | |
| | C5 | Check for both 3 or 6 for at least one roll (1) | any method e.g.<br>`if (roll == 3 or roll == 6)`<br>`if roll % 3 == 0` | |
| | C6 | booster incremented (1) | Must be indented under condition in C5.<br>Can award even if C5 not functioning correctly, but must be idea of only boosting on 3 and 6 | |
| | C7 | Correct final score calculated from 3 random rolls (1) | Even if not displayed<br>Does not need to use given variable (finalScore) to get this mark | **8** |

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# Type MS · Turn a flowchart or pseudocode into Python

## Answer 8 of 8 · Question 4 part c · 8 marks

2025 JUN · Paper 02 · extract 3 of 5 · find part c in this mark scheme extract



| | | | | |
|---|---|---|---|---|
| | C8 | Correct output strings including each roll score and final score with appropriate strings (1) | All three throws and final score as shown in figure 6.<br>`roll` and `finalScore` variables should match candidate solution<br>Allow sensible alternate strings e.g. Roll for Rolled or Total for Score – Ignore : (colon) | |

# Type MS · Turn a flowchart or pseudocode into Python

## Answer 8 of 8 · Question 4 part c · 8 marks

2025 JUN · Paper 02 · extract 4 of 5 · find part c in this mark scheme extract



```
# Q04c

# import libraries
import random

# program constants
SIDES = 6
THROWS = 3

# complete the lines to initialise the variables
roll = 0
subtotal = 0
booster = 1
finalScore = 0

# roll dice 3 times
for throw in range(THROWS):
    # display number rolled
    roll = random.randint(1,SIDES)
    print("Rolled:",roll)

    # update subtotal
    subtotal = subtotal + roll

    # check for roll of 3 or 6 and increment booster if needed
    if roll == 3 or roll == 6:
        booster = booster + 1

# calculate and output final score
finalScore = subtotal * booster
print("Final score:", finalScore)
```



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## Type MS · Turn a flowchart or pseudocode into Python

### Answer 8 of 8 · Question 4 part c · 8 marks

2025 JUN · Paper 02 · extract 5 of 5 · find part c in this mark scheme extract

Paper 2 – Application of Computational Thinking

| Question | mp | Answer | Additional guidance | Mark |
|---|---|---|---|---|